Donkey Car Simulator on Windows: Install Guide

Written for Fajar Laksono, from the working setup of October 2026.

Goal

Run the Donkey Car (Unity) simulator and control it from a Python script so telemetry (speed, cross-track error, position, images) can be collected.

The setup that works

Part	What was used
Where everything runs	Native Windows (simulator and Python script both on Windows)
Simulator	`donkey_sim.exe` (Windows build from the gym-donkeycar releases page)
Python	A virtual environment (`venv`) inside the project folder
gym-donkeycar	Installed from GitHub (version 1.3.1 or newer), not from PyPI
Gym library	Gymnasium 1.4.0
Connection	Script connects to the already-running sim on `127.0.0.1:9091`

The laptop is a Lenovo IdeaPad Gaming 3 (Ryzen 7 4800H, 16 GB RAM, 4 GB dedicated GPU plus Radeon integrated graphics). The simulator runs well on it when everything is on Windows.

Step 1: Download the simulator

1. Open https://github.com/tawnkramer/gym-donkeycar/releases

2. Download the Windows zip from the Assets section of the latest release (the name usually contains `Win`).
3. Unzip it to a simple folder, for example `D:\Project\DonkeySimWin\`.
4. Confirm `donkey_sim.exe` is inside. Double-click it once to check that the track menu appears.

Step 2: Create the project and virtual environment

In PowerShell:

```
cd D:\Project\Github\FajarLaksono\sopir
python -m venv venv
venv\Scripts\activate
python --version
```

The prompt should start with `(venv)`. Python 3.11 is the safest choice for compatibility. If `py -3.11` is available, you can use `py -3.11 -m venv venv` instead.

Always activate the venv in each new terminal before running scripts.

Step 3: Install gym-donkeycar from GitHub

Important: do **not** use plain `pip install gym-donkeycar`. PyPI gave version 1.0.13, which uses the old `gym` library and registers no environments in Gymnasium. Install from GitHub instead:

```
pip uninstall -y gym-donkeycar gym
pip install git+https://github.com/tawnkramer/gym-donkeycar
```

If `git` is not recognized, install Git for Windows, or use the zip instead:

```
pip install https://github.com/tawnkramer/gym-
donkeycar/archive/refs/heads/master.zip
```

If Gymnasium is missing, install it too:

```
pip install gymnasium
```

Step 4: Verify the install

Save as `check.py` and run `python check.py`:

```
import gymnasium as gym
import gym_donkeycar

print(gym_donkeycar.__file__)
print([k for k in gym.registry if "donkey" in k])
```

You should see a non-empty list with names like `donkey-generated-track-v0`. If the list is empty, the wrong gym-donkeycar version is installed. Check with `pip show gym-donkeycar`.

Step 5: Test script

Save as `test_sim.py`:

```
import gymnasium as gym
import gym_donkeycar

conf = {"exe_path": "remote", "host": "127.0.0.1", "port": 9091}
env = gym.make("donkey-generated-track-v0", conf=conf)
obs, info = env.reset()

for i in range(200):
    obs, reward, terminated, truncated, info =
env.step(env.action_space.sample())
    print(i, info.get("speed"), info.get("cte"))
    if terminated or truncated:
        obs, info = env.reset()

env.close()
```

`"exe_path": "remote"` means the script does not launch the sim. You start it yourself.

Step 6: Run it

1. Start `donkey_sim.exe` and leave it on the track menu.
2. In PowerShell, with the venv active:

```
python test_sim.py
```

The car should move in the sim window and the terminal should print speed and CTE values.

Optional: Log telemetry to CSV with images

Save as `log_telemetry.py`. The sim must already be running.

```
import csv
import os
import gymnasium as gym
import gym_donkeycar
from PIL import Image

conf = {"exe_path": "remote", "host": "127.0.0.1", "port": 9091}
env = gym.make("donkey-generated-track-v0", conf=conf)
obs, info = env.reset()
print("Available info keys:", list(info.keys()))

os.makedirs("data/images", exist_ok=True)
with open("data/telemetry.csv", "w", newline="") as f:
    w = csv.writer(f)
    w.writerow(["step", "img", "steering", "throttle", "speed", "cte", "x",
"y", "z", "hit"])

    for i in range(2000):
        action = env.action_space.sample()  # replace with your own
controller
        obs, reward, terminated, truncated, info = env.step(action)

        img_path = f"images/{i}.jpg"
        Image.fromarray(obs).save(f"data/{img_path}")
        x, y, z = info.get("pos", (None, None, None))
        w.writerow([i, img_path, action[0], action[1],
                    info.get("speed"), info.get("cte"), x, y, z,
info.get("hit")])
```

```
        if terminated or truncated:
            obs, info = env.reset()

env.close()
```

Telemetry fields depend on the simulator build. The official docs list steering angle, throttle, speed, image, hit, accelerometer and gyro values, pitch/roll/yaw, and node info, with position, velocity and CTE marked as training-only. Print the keys once to see what your build provides.

Troubleshooting log

These are the problems hit during setup and how each was fixed.

Symptom	Cause	Fix
`EnvironmentNameNotFound: donkey`	The conda env was never created	Create it, or use a venv instead
`donkey: command not found`	Only the gym wrapper was installed, not the full Donkeycar framework	Not needed for gym-only telemetry work
`CondaToSNonInteractiveError`	Conda requires accepting channel terms	Run the two `conda tos accept` commands from the error, or use `-c conda-forge --override-channels`
`ModuleNotFoundError: gymnasium` (in WSL)	`wsl python3` used the system Python, not the conda env	Run `conda activate donkey` inside WSL first
`NameNotFound: donkey-generated-track`	PyPI gym-donkeycar 1.0.13 is too old and does not register in Gymnasium	Install from GitHub (Step 3)
`ModuleNotFoundError: gymnasium` (in venv)	Gymnasium was not installed in the venv	`pip install gymnasium`

Symptom	Cause	Fix
Linux sim in WSL was very slow	WSLg likely used software rendering	Run the sim on Windows instead
Laptop crashed when script ran with sim on Windows and script in WSL	Not fully diagnosed; WSL combined with a leftover Linux sim and hybrid GPU are suspects	Running both on Windows was stable

Notes and lessons

- Running the sim and script both on Windows was stable and fast. Prefer this setup.
- If you try WSL again, run the sim on Windows with `"exe_path": "remote"` and use the Windows host IP (from `ip route | grep default`) as `"host"`. Make sure no Linux copy of the sim is still running (`pkill -f donkey_sim`).
- The laptop has hybrid graphics. If the sim ever misbehaves, set `donkey_sim.exe` to the high-performance GPU in Windows Settings, Display, Graphics, and keep NVIDIA and AMD drivers updated.
- Only one simulator instance should run at a time, because it listens on port 9091.

Next steps

- Replace the random actions with a controller (keyboard, joystick or a simple rule-based driver).
- Add timestamps to each logged row for consistent timing.
- If you want the full drive, train and test cycle, install the Donkeycar framework in a separate environment and follow the official guide at https://docs.donkeycar.com/guide/deep_learning/simulator/